

Digital Phenotyping based on a Mobile App Identifies Distinct and Overlapping Features in Children Diagnosed with Autism versus ADHD

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ABSTRACT

The high prevalence of autism calls for accessible and scalable technology-assisted screening tools. This will aid in early detection allowing timely access to services and supports. SenseToKnow, a mobile digital phenotyping app, showed potential in eliciting autism-related behaviors that can be automatically captured via computer vision analysis (CVA) in toddlers. Here, we present the capability of SenseToKnow in characterizing autism in school age children and showcase the robustness of the CVA features in interpreting distinct and overlapping behaviors with attention-deficit/hyperactive disorder (ADHD).

CCS CONCEPTS

• **Applied Computing** → Health informatics;; • **Human-centered computing** → Human computer interaction (HCI); Display and imagers.

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KEYWORDS

Autism, Digital Phenotyping, ADHD, Computer Vision

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1 INTRODUCTION

There is an increasing need for accessible and timely methods for early screening and diagnosis of developmental disorders in general and autism in particular [2, 9]. Signs of autism can be observed as early as 9 to 18 months, including decreased social attention, differences in facial expressions, and motor delays [1, 4, 10, 23]. While signs are present early, misdiagnoses often occur and the delay between the first clinical visit and diagnosis can take years [7, 8]. Earlier detection can lead to early intervention, while delayed intervention can lead to social and mental health issues [19, 20]. Autism often presents with cooccurring conditions such as attention-deficit/hyperactivity disorder (ADHD) [16]. This relationship and effect on behavior must be studied since behavior overlap can lead to misdiagnosis [22].

SenseToKnow (S2K), a mobile app we developed that uses computer vision analysis (CVA) and machine learning (ML) to capture autism-related behaviors has shown promise in screening for autism with high sensitivity and specificity in 16–40-month-old toddlers compared to neurotypical (NT) children and those with developmental delay [12, 24]. In this work, we administered S2K on a novel cohort, diverse in age and included participants with ADHD, autism, co-occurring ADHD and autism, and NT children.

2 RELATED WORK

Early autism screening is currently done using a caregiver questionnaire, such as the Modified Checklist for Autism in Toddlers-Revised with Follow-Up (M-CHAT-R/F) [25] and diagnosis via the Autism Diagnostic Observation Schedule-2 (ADOS-2) [18]. Due to lack of scalability and availability of trained therapists in using these tools, there is a recent focus on technology solutions. One such work utilizes eye-tracking and ML, showing high specificity but low sensitivity for autism diagnosis [26]. Another study with a sophisticated eye tracker showed a sensitivity of 71% and specificity of 80.7% [11]. Also, there are ongoing efforts in exploring the heterogeneous nature of the autism spectrum using brain imaging data to uncover distinct neurosubtypes within the autistic population [6, 17]. These techniques can be used to understand better how those with different diagnoses compare in behavioral features and how these features are present in those with cooccurring diagnoses.

3 METHODS

3.1 Digital Phenotyping App and Participants

The S2K app presents a series of ten social and nonsocial movies (refer to [24] for more details). Here, we explore the app's potential

for distinguishing autism in 40-100-month-old participants. The sample ($N = 181$) consisted of autistic children (AUT, $N = 51$), ADHD ($N = 48$), cooccurring autism and ADHD (AUT+ADHD, $N = 41$) and NT ($N = 41$).

3.1.1 App Administration and Analysis. S2K is presented on a tablet while participants watch the movies on a caregiver's lap. Simultaneously, we record the participant's using the front-facing camera. We identify and process the video, extracting child's landmarks and facial action units. We extract gaze features through a deep neural network to estimate social attention variables as in [3]. Using landmark time-series, we compute eyebrows and mouth movement complexities exploiting the multiscale entropy [14]. Using head pose, we calculate head movement rate, acceleration, and complexity [5, 13]. We compute blink rate using facial action units and compute how often the child is facing forward towards the screen [15]. In total we compute 17 features for social/nonsocial movies as mentioned in [24].

3.1.2 Computer Vision Analysis Validity Checklist. We built an automated CVA validity algorithm to evaluate the quality of participant's videos. Usable data were defined as (1) detecting facial landmarks for >50% of frames, (2) head pose angles less than 35°, and (3) distance from the device between 60-80 cm. Human validity coding and the CVA validity check showed high agreement (kappa $k=0.97$).

3.1.3 Privacy and Ethics. This research study's protocol was reviewed and approved by the DUHS Institutional Review Board. Written informed consent was obtained from all the caregivers. All participant's clinical information is anonymized, and videos are stored remotely in a secure location.

3.2 Statistical Tests

Group differences and effect of age are computed using Mann-Whitney U and ANCOVA, respectively. We divide the feature space into three sub-domains: (1) social attention (facing forward, blink rate, and gaze), (2) sensory motor (head movement-related features), and (3) facial expressions (eyebrows and mouth complexity). We perform latent profile analysis within each sub-domain using Gaussian Mixture Models (GMMs) to understand each sub-domain.

4 RESULTS

4.1 Administration Validity

The percentage of app administrations that were considered valid was high (95.5%) and is consistent across diagnoses.

4.2 Group Differences with ADHD and Relationship with Age

Comparing the AUT and AUT+ADHD groups, we find significant differences only in gaze ($p < .05$). Comparing ADHD and NT, we find significant differences in gaze and nonsocial blink rate ($p < .05$). However, comparing ADHD with AUT, 14 of 17 features have significant differences ($p < .05$). Combining AUT+ADHD and AUT participants and comparing to those with only ADHD, the same features remain significant ($p < .05$). Finally, comparing ADHD and AUT+ADHD show significant differences in 8 features, the most

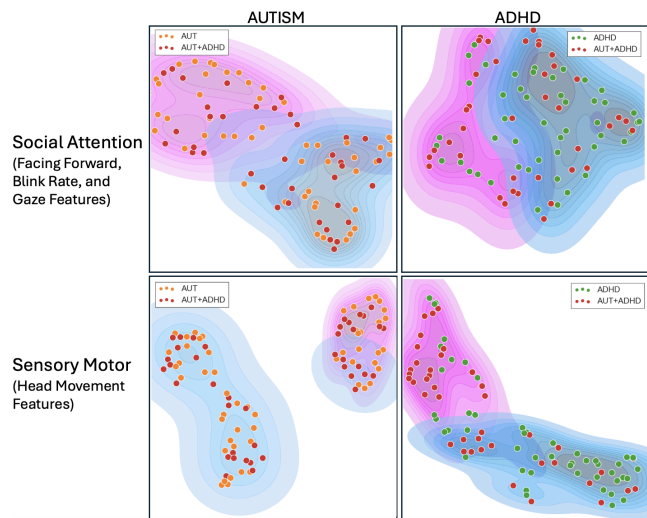


Figure 1: (a) Social attention (b) Sensory Motor mapped in latent space with UMAP [21]. Points colored by diagnosis and background colors by GMM cluster: pink (0) and blue (1).

significant being sensory-motor features ($p < .005$). With age as a covariate, all features show significant differences between diagnoses when combining AUT and AUT+ADHD and comparing to NT ($p < .001$). Comparing ADHD and NT groups, all nonsocial features and gaze silhouette score were significant ($p < .05$), accounting for age.

4.3 Latent Profile Analysis (LPA)

In both sensory-motor and social attention, two distinct clusters were formed, visualized in Figure 1. No clear subtypes were found for facial expressions. For this analysis, we considered AUT vs AUT+ADHD and ADHD vs AUT+ADHD and in each case two different clusters were formed. The separation in the sensory motor space results in significant cluster differences for all features ($p < 10^{-8}$) while for social attention, we find significant cluster differences in facing forward, blink and gaze features ($p < .0001$). For social attention, both clusters have a similar proportion of AUT and AUT+ADHD. However, in the ADHD case, cluster 0 (pink) displays significantly lower social attention features ($p < 10^{-6}$) and contains twice the number of ADHD (32 to 16), with a similar number of AUT+ADHD in each (20 to 19).

5 DISCUSSION

5.1 Robustness Across Age

Group differences remain consistent across age, showing the effectiveness of these features in the school-aged group.

5.2 Features Cater Towards Autism over ADHD

SenseToKnow features show substantial differences between autistic and non-autistic groups, even in presence of ADHD. The lack of significance between the ADHD and NT groups further indicates how the stimuli chosen and subsequent features analyzed are

specific towards understanding autism and its presentations. This should lend greater support to using S2K for screening autism.

5.3 LPA Reveals Distinctive and Interpretable Subspace

Distinct clusters were identified in both social attention and sensory motor. However, AUT and AUT+ADHD are equally distributed across clusters sharing similar behavior, stressing the specificity of autistic behavior.

6 CONCLUSION

This work presents the use of SenseToKnow app on a novel cohort to identify autism behaviors from a mixed sample of diagnoses of NT and ADHD across a wide age range. This tool needs only a tablet and the frontal camera and no other special equipment. Future work should concentrate on further validating these findings in a larger sample.

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